

MODEL QUESTION PAPER

TED (10) 5022

Fifth Semester Diploma Examination in Tool & Die Engineering Design of Jigs, Fixtures and Gauges

Time: 3 Hrs.

(Maximum Marks: 100)

PART – A

I Answer the following questions in one or two sentences (each carries 2 marks).

1. Define a Jig.
2. Write two essentials in design of Jigs.
3. Write the principle of location.
4. List materials for drill bushing.
5. Name three basic types of Limit Gauges.

PART – B

II Answer any five of the following. (each question carries 6 marks)

1. In mass production state the role of Jigs & Fixtures.
2. Explain the purpose of clamping.
3. Illustrate a quick action 'V' locator.
4. Explain Eccentric Clamping.
5. Sketch an angle plate Jig and name the parts.
6. Describe any one grinding fixture.
7. Illustrate Taylors Principle of Limit Gauges.

PART – C

Answer one full question from each module. (each question carries 15 marks)

Module – I

- III a) Describe a fool proofing method [7]
b) Summarise the defects in Jigs and Fixtures. [8]

OR

- IV a) Discuss Jigs and Fixtures. [7]
b) Explain degree of freedom. [8]

Module – II

- V a) Summarise the principle of location. [7]
 b) Illustrate a cam operated 'V' locator. [8]
- OR
- VI a) Summarise the principle of clamping. [7]
 b) Illustrate quarter twin screw clamping. [8]

Module III

- VII a) Describe various types of drill bushings. [7]
 b) Describe a Template drill Jig. [8]
- OR
- VIII a) Describe slip renewable bushing. [7]
 b) Illustrate a universal drill Jig. [8]

Module IV

- IX a) Explain the design consideration of clamps for fixtures. [7]
 b) Summarise the essentials of milling fixture. [8]
- OR
- X a) Describe any one type of welding fixture. [7]
 b) Illustrate the basic types of Limit Gauges. [8]
